

Omar Shawada

Production Planning Engineer

Location: Dakahlia, Egypt — Willing to Relocate

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PROFESSIONAL SUMMARY

Production Planning Engineer with 2+ years of experience in complex discrete manufacturing environments. Deep expertise in production scheduling, machine capacity planning, BOM-level constraint analysis, WIP control, and injection/assembly coordination. Built AI-powered scheduling tools that replaced static Excel plans with dynamic, constraint-aware systems — enabling real-time re-prioritization at machine level. Currently pursuing CPIM v8 (APICS). Known for translating floor-level production reality into executable daily and weekly plans.

KEY ACHIEVEMENTS

- **AI-Powered Production Scheduling System (AKL Production Planner):** Designed and built a dynamic scheduling web app. Input: Excel production plan with BOM data. Output: real-time daily Schedule + Machine Loading reports per machine. Supports live re-prioritization when any constraint changes (machine breakdown, order priority shift, capacity update) — eliminating hours of manual rescheduling every day.
- **Assembly Analyzer — BOM Constraint Engine:** Built an intelligent Excel tool that drills through every BOM level to identify exactly which component is blocking which finished product. Color-coded system: Green (simple) / Purple (deep multi-level BOM) / Blue (injection or procurement shortage) / Yellow (shared component blocking multiple products). Result: instant visibility into production blockers, enabling surgical scheduling decisions.
- **Gen AI Planning Agent — Ready-to-Assemble & Waiting-on-Injection Reports:** Designed a Custom GPT planning agent that reads production Excel data and automatically classifies every product into Ready to Assemble or Waiting on Injection. Delivers structured reports used directly in daily scheduling decisions.
- **Production Visibility Dashboard:** Built a real-time dashboard covering machine status, WIP levels, component shortages, production stoppages, and order progress — giving the planning team instant floor visibility without manual data chasing.

PROFESSIONAL EXPERIENCE

Production Planning Engineer | AKL Auto Feeding Industries | Apr 2025 – Present

Production Scheduling & Capacity Planning

- Develop daily and weekly production schedules for Injection and Assembly lines, balancing machine capacity, operator availability, mold changeover times, and order priorities.
- Manage machine loading across injection molding cells — allocating jobs per machine based on cycle time, mold compatibility, and capacity constraints.
- Monitor WIP levels across production stages; flag excess WIP and production imbalances before they cause downstream stoppages.
- Re-prioritize production orders in real time when machine breakdowns, material shortages, or urgent customer orders occur — maintaining floor execution without losing OTIF targets.
- Coordinate with Production Supervisors and Shift Leaders to align daily schedule execution with floor reality.

BOM Analysis & Constraint Management

- Analyze multi-level BOMs to identify component availability gaps before releasing production orders to the floor.
- Use Assembly Analyzer to map each finished product stoppage to its exact BOM level, injection component, or raw material — enabling targeted corrective actions per production stage.
- Classify production orders daily into Ready-to-Assemble vs. Waiting-on-Injection categories, providing clear priorities for Assembly and Injection supervisors.
- Coordinate with Injection team on mold scheduling and cycle time planning to ensure upstream components feed Assembly on time.

SAP Production Order Management

- Create, release, and manage production orders in SAP ByDesign (PP module) aligned with the master production schedule.
- Manage sales allocation orders in SAP; ensure high-priority customer orders are fulfilled first within available capacity.
- Track production order status and flag deviations between planned vs. actual output for daily review.

Reporting & Continuous Improvement

- Generate daily production status reports covering output, stoppages, WIP status, and BOM-level blockers — distributed to Production, Planning, and Management.
- Prepare weekly capacity utilization analysis per machine and production cell to support medium-term planning decisions.
- Identify recurring production bottlenecks and propose scheduling rule improvements to increase throughput and reduce idle time.

Sheet Metal Designer & Workshop Manager | Independent Workshop | May 2024 – Apr 2025

- Managed full production scheduling for laser cutting and bending operations — allocating jobs per machine based on material type, thickness, and sequence optimization.
- Applied Lean Manufacturing principles: 5S implementation, part nesting optimization to reduce material waste, micro-joint techniques to reduce bending time and cost.
- Built AppSheet mobile apps to digitize machine-level production data collection — enabling real-time job tracking and output reporting across all machines.
- Translated client requirements into SolidWorks and AutoCAD designs; generated production-ready DXF files optimized for laser cutting sequences.

TECHNICAL SKILLS

Production Planning: Production Scheduling, Machine Loading, Capacity Planning, WIP Control, MRP, OTIF, Cycle Time

BOM & Manufacturing: Multi-level BOM Analysis, Injection Scheduling, Assembly Coordination, Mold Planning, Constraint Analysis

ERP & Data: SAP ByDesign (PP, MM, SD), Excel, Power Query, Data Visualization, SQL (Beginner)

AI & Automation: AI Scheduling System, Custom GPT Planning Agent, Claude AI, Vibe Coding, AppSheet

Lean & CI: 5S, Kaizen, Waste Reduction, Visual Management, Process Documentation

EDUCATION

B.Sc. Mechanical Power Engineering — Mansoura University

Accumulative Grade: Good (67.33%) | 2017 – 2022

CERTIFICATIONS & DEVELOPMENT

- CPIM v8 – Preparation Program (APICS) — In Progress
- Supply Chain Digital Transformation — SQL, AI Agents, Custom GPT, Data Analysis
- SAP Introduction — PP, MM, SD Modules
- Data Analysis & Visualization — Excel, SQL
- AI Copilot: AI Productivity & Business Automation
- Global Industrial Management — Lean, TPM, 5S, OEE, KPIs (45 Hours)
- Process Improvement Foundations — LinkedIn Learning

LANGUAGES

Arabic: Native | English: Professional working proficiency (reading, writing, speaking)